

Summary

Results-driven roboticist and technical program manager with experience leading cross-functional engineering teams in fast-paced startups. I have over a decade of real-world experience deploying robotic systems in the field, including several years as a software engineer on safety-critical stacks. I lead autonomous systems projects from conception to deployment, balancing evolving requirements with proactive risk mitigation in iterative development cycles.

Employment

Waymo – Release Manager

April 2026 - Present | Remote

Software release manager for a multi-OEM, multi-vehicle-type autonomous vehicle fleet

- Managed software release process across a multi-configuration multi-national fleet
- Developed release, triage, and operational process improvements across a complex av software stack
- Established release metrics and evaluation infrastructure, reducing release labor by 25%
- Built a release hub unifying disconnected systems into one operational view improving visibility
- Managed release programs building cross-functional consensus to improve release process
- Developed an LLM-powered agentic system that automated parts of the release workflow reducing manual release blocker resolution time.

Thirdwave Automation – Technical Lead Manager

2020 - 2026 | Remote

Semi-autonomous forklifts operated as a multi-vehicle fleet

- Led a 7-person engineering operations team and served as company-wide Release TPM
- Team owned Testing, Release Management, Logging, Data, Triage, Calibration, and Sensor Verification
- Aligned teams on release priorities and partnered with subject matter experts to harden releases
- Architected testing and validation for multiple vehicle types, scheduling projects to mitigate risk
- Developed a safety-critical obstacle detector, enabling ANSI B56 certification
- Led a cross-company tiger team that delivered 3x pallet moves per hour in two months
- Developed testing/CI for the data pipeline, improving reliability by an order of magnitude

Embark Trucks – Senior Robotics Engineer

2018 - 2020 | San Francisco, CA

Autonomous semi trucks on highways; company IPO at \$5B market cap

- Joined as the 16th employee and contributed to company growth to over 200 employees
- Code owner for calibration, lane estimation, and scene generator; organized projects and managed junior engineers
- Developed Vision Map Fusion, extending planning field of view by 600% with a map-fused prior
- Developed pre- and post-trip auto-calibration, reducing vehicle downtime by ~15%

Apple – Special Projects Group

2016 - 2018 | Sunnyvale, CA

Project Titan, Apple's autonomous car program

- Built a ground-truth mapping pipeline fusing aerial, survey, and on-vehicle logs so localization metrics could be reported for each release
- Implemented a LiDAR sensor model in Unity, enabling off-vehicle validation and testing
- Created a low-cost modular distributed embedded traffic monitoring system for stoplight data collection

Education and Internships

Carnegie Mellon University, Pittsburgh, PA

- B.S. in Electrical Computer Engineering (ECE) 2015
- M.S. in Electrical Computer Engineering (ECE) 2016
 - Minors in Computer Science, Robotics, and Business Administration
 - Intern, Amazon Prime Air Summer 2015
 - Intern, Volkswagen Electronics Research Lab Summer 2014

Research Assistant, CMU Field Robotics Center

2011 - 2016

- Assisted with software and hardware maintenance/development of 10+ robots
- Built multi-robot co-localization algorithms in low infrastructure environment

Teaching Assistant, CMU Introduction to Robotics

2013 - 2016

- 2016 lead teaching assistant, managed team of 12 TAs, wrote exams, taught select lectures
- Designed new Bayes filter localization lab, where students solve the lost robot problem

Activities

Robotic Buggy

2013 - 2016

Created an autonomous robot to compete in a traditionally human-only gravity race at CMU

- Led a cross-functional team of eight engineers, planning all development and testing
- Made feature trade-offs to guarantee success in the robot's single allotted time slot
- Developed a scalable real time architecture for mapping, path planning, and localization
- Wrote motion model and observation model for GPS, IMU, encoders, cameras, ...
- Built computer vision road lane and feature detectors to help extract robot's state

Distinctions

- Patents US9823089 and US10302452B1 for online drone calibration
- 1st place 2017 Apple maps emerging technology reinforcement learning competition
- Computer technician: diagnosed and repaired 650+ laptop computers for school students (2009 - 2011)
- Black Belt, Tang Soo Do Karate
- Won "Coolest Robot" award for skyscraper window cleaner, CMU Mechatronics 2016
- FIRST Robotics 3 years high school participant, 2 years mentor

Skills and Experience

Release & CI/CD: Release Management*, Jira*, Buildkite, Bazel*, Docker*, Nomad*, Kubernetes, Terraform, Ansible, SQL*, Team Management*, Project Planning*

AV & Safety: Autonomous Vehicles*, Calibration*, Sensor Fusion*, Simulation*, Localization, Mapping

Software: Python*, C, C++, JavaScript, React, Java, ROS

Hardware: 3D printing, mill, lathe, carbon fiber layup, electronic assembly, SolidWorks/CAD, forklift certified

*Used in current role